

EXPERIMENT: PLACEMENT STATISTICS DASHBOARD VISUALIZATION

AIM:

To create data visualizations such as pie charts and bar graphs to analyze placement statistics using JavaScript and Chart.js.

OBJECTIVE:

- To understand data visualization techniques
- To represent placement data using charts
- To use Chart.js for creating interactive graphs
- To display values clearly inside a pie chart

INTRODUCTION:

Data Visualization is the graphical representation of data using charts and graphs. In this experiment, a Placement Statistics Dashboard is developed to visualize placement data such as student placements across different sectors like IT Company, Core Company, Startup, and Higher Studies.

The dashboard also compares placement counts department-wise using a bar graph for departments such as CSE, IT, ECE, and EEE. This helps in understanding placement trends and improves decision-making through graphical representation.

PROCEDURE:

Step 1: Create HTML File (index.html)

- Create a basic HTML structure
- Add canvas elements for Pie Chart and Bar Graph
- Include Chart.js library and datalabels plugin

Step 2: Create JavaScript File (script.js)

- Define placement data
- Configure pie chart with values inside the chart
- Configure bar graph for department-wise placement statistics

Step 3: Run the Project

- Open index.html in a browser
- Visualize the charts generated using Chart.js

CODE:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Placement Statistics Dashboard</title>
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script src="https://cdn.jsdelivr.net/npm/chartjs-plugin-datalabels"></script>
<style>
body {
font-family: Arial, sans-serif;
text-align: center;
background: #f5f5f5;
}

.container {
width: 80%;
margin: auto;
}

canvas {
margin: 30px 0;
background: white;
padding: 20px;
border-radius: 10px;
box-shadow: 0 0 10px rgba(0,0,0,0.1);
}
</style>
</head>
<body>

<div class="container">
<h1>Placement Statistics Dashboard</h1>

<h2>Placement Distribution</h2>
<canvas id="pieChart"></canvas>
```

```
<h2>Department-wise Placements</h2>
<canvas id="barChart"></canvas>
</div>
```

```
<script src="script.js"></script>
```

```
</body>
</html>
```

script.js

```
const pieCtx = document.getElementById('pieChart').getContext('2d');
```

```
new Chart(pieCtx, {
  type: 'pie',
  data: {
    labels: ['IT Company', 'Core Company', 'Startup', 'Higher Studies'],
    datasets: [{
      data: [40, 25, 20, 15]
    }]
  },
  options: {
    plugins: {
      legend: {
        position: 'bottom'
      },
      datalabels: {
        color: '#fff',
        font: {
          weight: 'bold',
          size: 14
        },
        formatter: (value) => value
      }
    }
  },
  plugins: [ChartDataLabels]
});
```

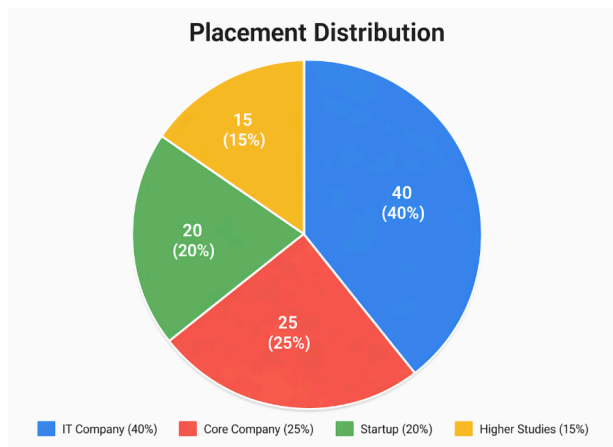
```
const barCtx = document.getElementById('barChart').getContext('2d');
```

```
new Chart(barCtx, {
  type: 'bar',
  data: {
    labels: ['CSE', 'IT', 'ECE', 'EEE'],
    datasets: [{
```

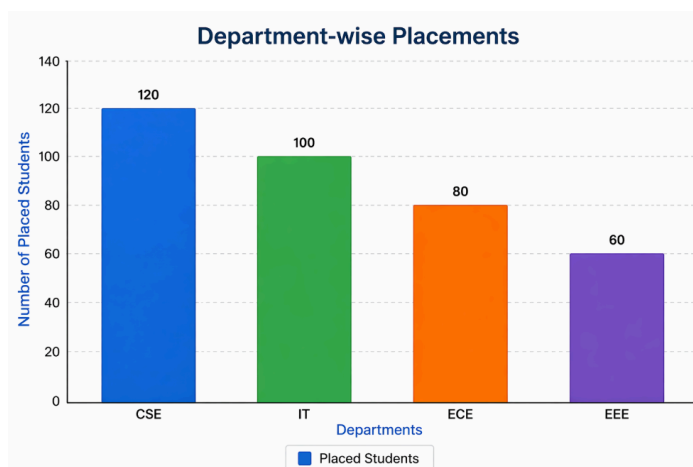
```
label: 'Placed Students',
data: [120, 100, 80, 60]
}],
options: {
responsive: true,
plugins: {
legend: {
display: true
}}}
});
```

OUTPUT:

Pie Chart showing placement distribution



Bar Graph comparing department-wise placements



RESULT:

Thus, the Placement Statistics Dashboard visualization was successfully created using JavaScript and Chart.js with values displayed clearly inside the pie chart and department-wise comparison shown using a bar graph.